Classification of Pediatric Brain Tumor Localization

Multi-Omic Integration (Gene Expression + CGH) in the $p \gg n$ Regime

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Presentation Outline

1. Introduction and Problem Formulation
2. Exploratory Data Analysis (EDA)
3. Formalism and Methodology
4. Results
5. Stability and Reproducibility
6. Literature Review

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Clinical Context and Problem Formulation

The Problem

Predict the **anatomical localization** of a pediatric glioma strictly from the tumor's **molecular signature**. Localization strongly dictates surgical resectability, radiotherapy protocols, and overall patient prognosis.

Three anatomical localizations define three target classes:

- **cort**: cortical / hemispheric tumors;
- **dipg**: diffuse intrinsic pontine glioma, located in the pons (inoperable, highly unfavorable prognosis);
- **midl**: midline tumors outside the pons (**rare class**).

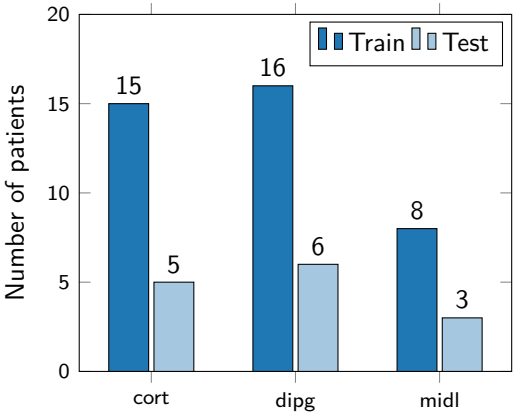
The Dataset

Two molecular blocks:

- **GE**: gene expression: **15,702** variables (primary discriminant signal);
- **CGH**: chromosomal gains / losses: **1,229** variables (weak signal in isolation).

Sample size: $n = 53$ patients: **39** in training and **14** in test.

	cort	**dipg**	**midl**
Train	15	16	8
Test	5	6	3



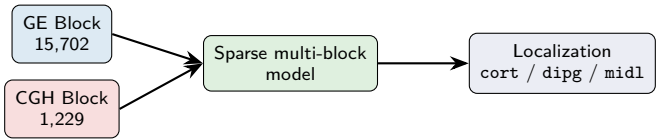
Class distribution: midl is substantially under-represented.

The Statistical Challenge: Three Bottlenecks

1. **Extreme $p \gg n$ regime**: $p = 16{,}931$ variables for $n = 39$ patients, yielding $\frac{p}{n} \simeq 434$. The Gram matrix $X^\top X$ has rank $\leq n - 1 = 38$, hence is *massively singular*: any unregularized estimator is non-identifiable.
2. **Two heterogeneous and imbalanced blocks**: GE accounts for 92.7% of variables; multi-block fusion must avoid marginalizing CGH.
3. **Rare class midl**: 8 patients in train, 3 in test. Each error shifts its recall in steps of $1/3 \simeq 0.33$.

Methodological Responses

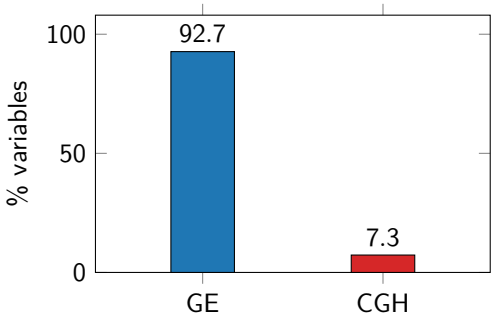
- ℓ_1/ℓ_2 regularization + sparse feature selection;
- Block scaling + supervised latent projection (SGCCA);
- Likelihood weighting + **balanced accuracy**.



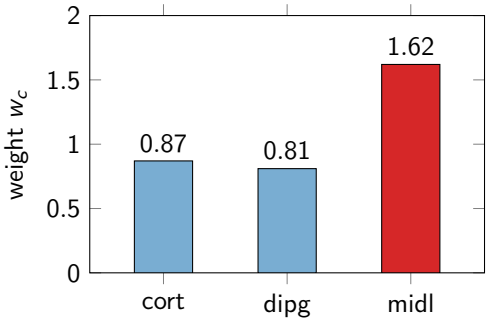
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GE Block Domination and Class Weighting

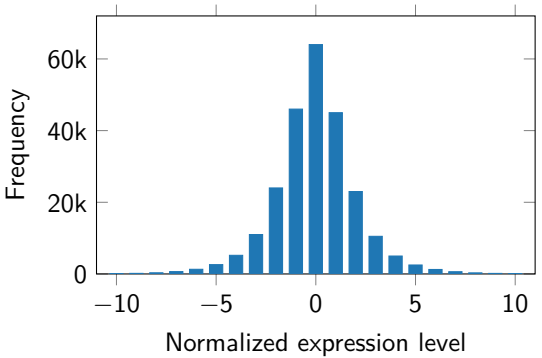


Inertia $\propto$ number of variables: raw concatenation is $>$ 90% dominated by GE.



Inverse prevalence weights $w_c = n/(K\, n_c)$: a midl patient weighs $\simeq 1.6\times$ an average patient.

Data Quality: Pre-Normalized GE Expression Levels

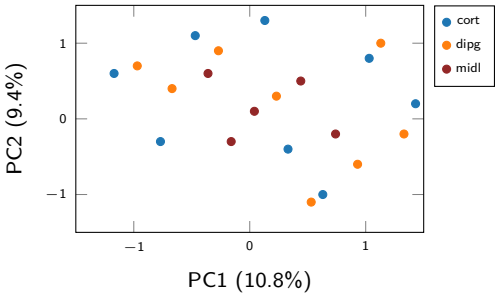


Global distribution of the $39 \times 15{,}702$ expression values:

- Quasi-symmetric, centered (median $= 0.03$), bounded (-10.43 to 9.87);
- Pre-normalized data (log-ratios) $\Rightarrow$ no non-linear transformation (log, Box-Cox) required; simple z-score standardization suffices.

Imputation via training median (carried over to test): robust to outliers, strictly leak-free.

Unsupervised PCA Fails to Separate Classes



- PC1+PC2 capture only $\simeq 20\%$ of total variance for GE ($\simeq 27\%$ for CGH);
- The three point clouds **overlap heavily**; `midl` forms *no distinct cluster*.

Consequence

Unsupervised dimension reduction (PCA, Sparse PCA) projects variance along non-discriminative axes $\Rightarrow$ failure of Sparse PCA + LogReg pipelines. **A supervised projection is mandatory** (SGCCA).

Real but Fragile Univariate Signal, High Redundancy

***t*-tests (Welch) `midl` vs. rest** (Welch due to unequal variances, $n_{\text{midl}} = 8$):

- **GE**: a few genes reach $|t| \simeq 4$–5 (e.g., `A_32_P85999`) $\Rightarrow$ univariate signal exists;
- **CGH**: statistics plateau around $|t| \simeq 3$, substantially weaker signal.

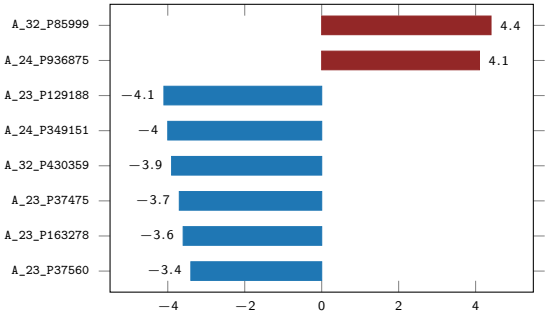
Strong intra-block redundancy (correlated feature clusters, especially in CGH):

- Justifies **Elastic-Net** penalization

Univariate $\neq$ Robust Multivariate

With $p \gg n$ and $n_{\text{midl}} = 8$, these t-statistics are themselves noisy (a single patient can alter rankings) and do **not** account for multiplicity ($\sim 17{,}000$ tests). This is an exploratory diagnostic, not valid feature selection $\Rightarrow$ selection is delegated to penalized methods *intra-CV*.

Univariate Signal around `midl`: Top Features (Welch's t)



Welch's t (`midl` vs. rest)

- **GE**: a few genes reach $|t| \simeq 4$–5; red = over-expressed in `midl` (e.g., `A_32_P85999`), with the majority under-expressed;
- **CGH**: statistics plateau around $|t| \simeq 3$, consistent with its near-total exclusion by sparse models.

Metric Selection: Balanced Accuracy

Raw *accuracy* implicitly weights by class prevalence and would mask complete failure on `midl` (a trivial classifier never predicting midl already reaches $\simeq 11/14$).

We adopt **macro-recall** (= balanced accuracy):

$$\mathrm{BA}(\hat{f}) = \frac{1}{|\mathcal{Y}|} \sum_{c \in \mathcal{Y}} \underbrace{\frac{|\{i : Y_i = c, \hat{f}(X_i) = c\}|}{|\{i : Y_i = c\}|}}_{\mathrm{recall}_c}$$

Equal weight assigned to all three classes, irrespective of prevalence.

An Example

For a classifier that *never* predicts `midl`:

$$\text{macro-recall} = \frac{1}{3}\left(\frac{5}{5} + \frac{6}{6} + \frac{0}{3}\right) = 0.667$$

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Probabilistic Framework and Bayes Classifier

3-class classification: (X_i, Y_i) i.i.d. $\sim P$, $X \in \mathbb{R}^p$, $Y \in \{1, 2, 3\}$. Risk of a classifier f under loss ℓ:

$$R(f) = \mathbb{E}_{(X,Y)\sim P}[\ell(f(X), Y)], \qquad f^\star = \arg\min_f R(f).$$

Under 0–1 Loss

With $\eta_c(x) = \mathbb{P}[Y = c \mid X = x]$, the Bayes optimal predictor is

$$f^\star(x) = \arg\max_{c \in \mathcal{Y}} \eta_c(x).$$

Since 0–1 loss is neither continuous nor differentiable, we optimize a convex, smooth **surrogate loss**: cross-entropy.

Remark: Under an *asymmetric* loss (penalizing errors on `midl` more heavily), the Bayes predictor is no longer the argmax of η_c: this motivates cost-sensitive class weighting.

Multinomial Logistic Regression

Softmax model (with identifiability constraint $\beta_3 = 0$):

$$\eta_c^\beta(x) = \mathbb{P}_\beta[Y = c \mid X = x] = \frac{e^{\beta_c^\top x}}{\sum_{k=1}^{3} e^{\beta_k^\top x}}.$$

Maximizing log-likelihood $\equiv$ minimizing empirical **cross-entropy**:

$$\hat{R}_n(\beta) = -\frac{1}{n}\sum_{i=1}^{n} \log \eta_{Y_i}^\beta(X_i), \qquad \nabla_{\beta_c} \hat{R}_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} \left(\eta_c^\beta(X_i) - \mathbf{1}_{\{Y_i = c\}}\right) X_i.$$

- **Convex and differentiable** objective; no closed-form solution $\Rightarrow$ Newton-Raphson method.
- **Linear** decision boundaries: $\hat{y} = \arg\max_c \hat{\beta}_c^\top x$.

Weighted Likelihood: Inverse Prevalence Weights

The uniform empirical risk $\hat{R}_n$ is dominated by majority classes. We align the objective with balanced accuracy via

$$\hat{R}_n^w(\beta) = -\frac{1}{n}\sum_{i=1}^{n} w_{Y_i} \log \eta_{Y_i}^{\beta}(X_i).$$

Weights

Equalizing class contributions ($w_c n_c = \text{const}$) subject to normalization ($\sum_c w_c n_c = n$) yields

$$w_c = \frac{n}{K\, n_c}, \qquad w_{\texttt{cort}} = 0.87, \quad w_{\texttt{dipg}} = 0.81, \quad w_{\texttt{midl}} = 1.62.$$

The weighted gradient scales `midl` residuals by $w_{\texttt{midl}}$. This is theoretically more sound than synthetic oversampling like `SMOTE` (evaluated, inconclusive: synthesizing pseudo-patients in dimension 16,931 induces severe instability). Convexity is strictly preserved.

Elastic-Net Regularization and GLM Framework

Elastic-Net to control complexity in the $p \gg n$ regime:

$$\min_{\beta} \left\{ \hat{R}_n(\beta) + \lambda\rho\|\beta\|_1 + \frac{\lambda(1-\rho)}{2}\|\beta\|_2^2 \right\}$$

- ℓ_1 (Lasso) $\Rightarrow$ feature **selection**
- ℓ_2 (Ridge) $\Rightarrow$ **stability** under collinear features
- Non-differentiable ℓ_1 $\Rightarrow$ proximal optimization (soft-thresholding).

GLM

Logistic regression is the canonical *GLM*: $Y \mid X \sim$ Categorical, logit link, $g(\mu_i) = x_i^\top \beta$. Log-likelihood is concave $\Rightarrow$ unique MLE obtained via Newton (iteratively reweighted least squares, IRLS).

Linear Discriminant Analysis (LDA)

Assumptions: conditional normality $X \mid Y = c \sim \mathcal{N}_p(\mu_c, \Sigma)$ and **homoscedasticity** ($\Sigma_c = \Sigma$). The discriminant function linearizes:

$$\delta_c(x) = \underbrace{(\Sigma^{-1}\mu_c)^\top}_{w_c^\top} x - \underbrace{\left(\tfrac{1}{2}\mu_c^\top \Sigma^{-1}\mu_c - \log \pi_c\right)}_{b_c}, \qquad f^\star(x) = \arg\max_c w_c^\top x - b_c.$$

Estimators: class priors $\hat{\pi}_c = \frac{n_c}{n}$, class centroids $\hat{\mu}_c$, and pooled covariance

$$\hat{\Sigma} = \frac{1}{n-K} \sum_c \sum_i (X_i - \hat{\mu}_c)(X_i - \hat{\mu}_c)^\top \mathbf{1}_{Y_i = c}.$$

Problem in $p \gg n$

$\hat{\Sigma}$ ($16{,}931 \times 16{,}931$) is **singular (non-invertible)**. LDA is therefore never applied directly to raw features, but rather to the low-dimensional *supervised latent components* obtained via SGCCA.

Cross-Validation: Stratified, Repeated, Nested

With $n = 39$, any single train/validation split exhibits extreme variance.

- **Stratification**: every fold preserves exact class proportions (preventing folds with zero `midl`);
- **Repetition**: $K \times R = 7 \times 3 = 21$ folds to reduce the variance of $\hat{R}_{\text{CV}}$;
- **Metric**: balanced accuracy optimized and reported throughout.

Nested CV

- *Outer loop*: estimates $\hat{R}_{\text{CV}}$;
- *Inner loop*: selects $\hat{\lambda} = \arg\min_{\lambda} \hat{R}_{\text{in}}(\lambda)$ strictly on the fold training split.

$\Rightarrow \hat{\lambda}$ is tuned without ever seeing test folds: zero **data leakage**. All feature selection is performed *strictly within* cross-validation folds.

Multi-Block Fusion: From Naive Baselines to Cooperative Learning

Two blocks: $X \in \mathbb{R}^{n \times p_{\text{GE}}}$ and $Z \in \mathbb{R}^{n \times p_{\text{CGH}}}$.

- **Early fusion**: concatenate $[X \mid Z] \Rightarrow$ GE completely dominates; CGH is marginalized;
- **Late fusion**: separate model per block, then $\omega f_X + (1-\omega) f_Z \Rightarrow$ discards inter-block feature interactions.

Cooperative Learning (Ding & Tibshirani, PNAS 2022)

Continuous interpolation between early and late fusion via an **agreement penalty** $\rho \geq 0$ on predicted logits:

$$J(\theta_x, \theta_z) = \hat{R}_{\text{CE}}(\theta_x, \theta_z) + \underbrace{\frac{\rho}{2} \|X\theta_x - Z\theta_z\|_F^2}_{\text{agreement}} + P_{\text{Elastic-Net}}(\theta_x, \theta_z).$$

$\rho = 0$: early fusion; $\rho \to \infty$: late fusion. Equivalent to Elastic-Net optimization on *augmented* feature data $(\tilde{X}, \tilde{z})$.

SGCCA: The Supervised Latent Projection Framework

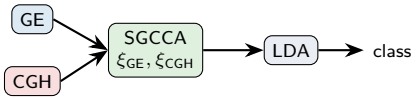
Sparse Generalized CCA (Tenenhaus *et al.*): sparse components $y_j = X_j a_j$ maximizing inter-block covariance *across connected blocks*:

$$\max_{a_1,\dots,a_J} \sum_{j,k} c_{jk} g\left(\operatorname{cov}(X_j a_j, X_k a_k)\right) \quad \text{s.t.} \quad \|a_j\|_2 \le 1,\ \|a_j\|_1 \le s_j.$$

- **3 blocks**: GE, CGH, and dummy label matrix Y as a **supervised pivot**;
- Design matrix: each molecular block is linked directly to Y, rather than linking GE–CGH directly

$$C = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix};$$

- ℓ_1 budget $\Rightarrow$ variable selection; ℓ_2 unit sphere $\Rightarrow$ stability;
- Block scaling $\tilde{X}_j = X_j / \sqrt{p_j}$ to balance block inertia.



Latent components feed into LDA in low-dimensional space, naturally bypassing the singularity of $\hat{\Sigma}$.

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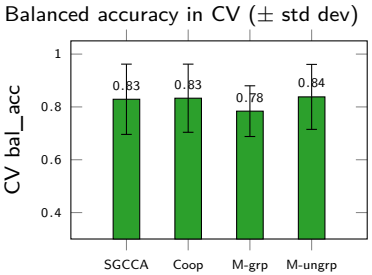
Feature Selection and Pipeline Benchmark

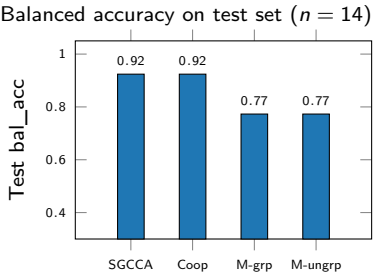
Four supervised sparse pipelines evaluated under an identical protocol ($7 \times 3 = 21$ stratified folds, balanced accuracy, held-out test set $n = 14$). SGCCA selects **68 GE + 11 CGH** ($s_1 = 0.051$, $s_2 = 0.067$).

Method	**CV bal_acc**	**Test bal_acc**	**`midl` test**	**N vars**
Multinomial Lasso *grouped*	0.784 ± 0.096	0.773	0/3	27 GE
Multinomial Lasso *ungrouped*	0.838 ± 0.123	0.773	0/3	27 GE
Cooperative Lasso OvR ($\rho = 0$)	0.833 ± 0.129	0.924	2/3	42 GE
SGCCA + LDA	0.829 ± 0.133	0.924	2/3	68 GE + 11 CGH

- Only *One-vs-Rest* or *supervised latent space* pipelines recover the `midl` class (2/3);
- Switching the ℓ_1 penalty from *grouped* $\rightarrow$ *ungrouped* gains $+0.054$ in CV (at the sole cost of a command-line flag), but yields identical test performance.

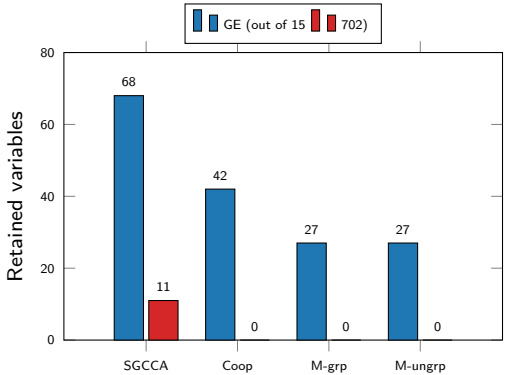
Benchmark Performance





In CV, three methods plateau around $\simeq 0.83$; on the test set, only **SGCCA + LDA** and **Cooperative OvR** achieve 0.924.

Effective Sparsity: Retained Variables per Block



All purely *data-driven* methods spontaneously zero out the entire CGH block.

Only **SGCCA** retains 11 CGH features: strictly because sparsity constraints s_{CGH} are enforced block-by-block by design.

Highly sparse and interpretable models (≤ 68 features out of $\sim 17{,}000$).

Cooperative Learning: An Instructive Instability

Under `multiview` (built on `glmnet`), the algorithm **diverges for any** $\rho > 0$ under the OvR formulation (`algorithm did not converge`). Two structural root causes:

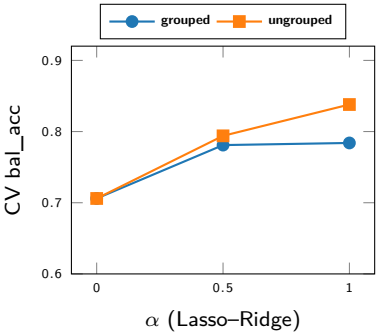
- Newton weights $w_i = \hat{p}_i(1 - \hat{p}_i) \to 0$ on the rare class $\Rightarrow X^\top W X$ becomes nearly singular;
- The agreement penalty injects $\rho X^\top X$, amplifying already severe collinearity by a factor of $(1 + \rho)$.

Structural Insight

The ℓ_1 penalty promotes *sparsity* while the agreement penalty enforces *view coupling*: antagonistic forces when GE and CGH share minimal common discriminant signal. Only $\rho = 0$ is viable (collapsing into standard OvR Lasso). The *multinomial* formulation converges ($\rho^\star = 0.10$), but at lower balanced accuracy.

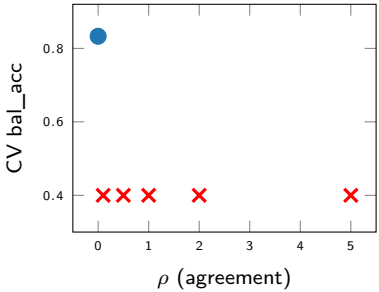
Hyperparameter Sensitivity

Multinomial Lasso: α sweep



At $\alpha = 1$, switching *grouped* $\rightarrow$ *ungrouped*: $+0.054$.

Cooperative OvR: ρ sweep



Red crosses: solver non-convergence (`multiview`/`glmnet`); only $\rho = 0$ is viable.

Multinomial vs. OvR: The Rare Class Bottleneck

Identical loss formulations yield divergent behaviors in $p \gg n$ under severe class imbalance.

SGCCA + LDA / Cooperative OvR

	p.cort	p.dipg	p.midl
t.cort	5	0	0
t.dipg	0	6	0
t.midl	0	1	**2**

bal_acc = 0.924: `midl` 2/3

Multinomial Lasso (grouped/ungrouped)

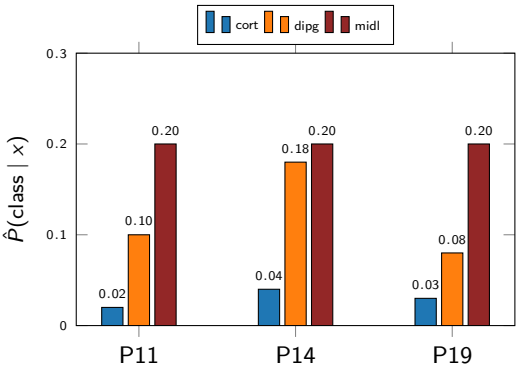
	p.cort	p.dipg	p.midl
t.cort	5	0	0
t.dipg	0	6	0
t.midl	1	2	0

bal_acc = 0.773: `midl` 0/3

The OvR "Safety Net"

In OvR, a `midl`-vs-rest sub-classifier with zero coefficients maintains an *intercept* calibrated to training prevalence: $\hat{P}_{\texttt{midl}} \simeq \sigma(\log(8/31)) \simeq 0.205$ *constant*. When `cort` and `dipg` reject an ambiguous patient, this baseline probability wins the arg max decision (e.g., P11, P19). Conversely, the multinomial sum-to-one constraint $\sum_c P_c = 1$ pulls this unallocated probability mass toward `dipg`.

Rescuing midl: Predicted Class Probabilities (Cooperative OvR)



For the 3 true test `midl` patients, OvR yields an almost constant baseline probability

$$\hat{P}_{\texttt{midl}} \simeq \sigma(\log(8/31)) \simeq 0.205.$$

With `cort` and `dipg` rejected, this baseline wins via **argmax** (P11, P19).

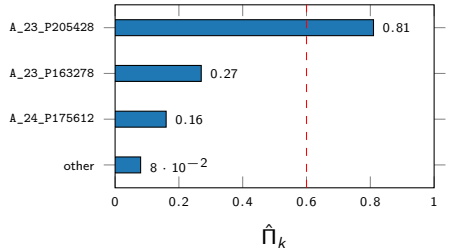
Multinomial models, bound by $\sum_c P_c = 1$, drain this mass into `dipg` $\Rightarrow$ `midl` recall $= 0/3$.

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Stability of Feature Selection

In the $p \gg n$ regime, selected feature sets exhibit variance under perturbations. We quantify stability via **bootstrap resampling** (Efron, 1979): selection frequency $\hat{\Pi}_k = \frac{1}{B} \sum_b \mathbf{1}_{\{a_k^{(b)} \neq 0\}}$, stable set $= \{k : \hat{\Pi}_k \geq \pi_{\text{thr}}\}$ (Meinshausen & Bühlmann, 2010; $\pi_{\text{thr}} = 0.6$, $B = 200$).



OvR Lasso: one stable top gene (A_23_P205428, $\hat{\Pi} \simeq 0.81$).

- A **core set of GE genes** is consistently selected;
- **SGCCA**: *more* stable core (3 genes $> \pi_{\text{thr}}$, sharing the same top gene as Lasso $\Rightarrow$ cross-method concordance);
- The **exclusion of CGH** by data-driven models is *remarkably stable*: a genuine finding, not an artifact.

Is the Signal Genuine? Permutation Testing

Permutation test (Ojala & Garriga, 2010):

- H_0: no statistical association between features and labels (expected BA $= 1/3$);
- Permute labels, retrain the entire pipeline from scratch, $B = 500$ repetitions;
- **p-value** = fraction of permutations matching or exceeding observed score.

Result

All models: **p-value** $= 0.002$. We reject H_0: predictive signal is statistically significant and *not attributable to chance*, despite $p \gg n$.

Reproducibility

Fixed random seeds (CV splits, bootstrap, solver initializations) $\Rightarrow$ exactly reproducible numbers.

Robust vs. Fragile

Robust: primary signal driven by GE; CGH uninformative; clear `cort`/`dipg` separation; global significance.

Fragile: fine ranking among top methods; exact `midl` recall (hinges on 1–2 test patients); exact gene list (stable core, volatile periphery).

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Literature Review: Foundations and Positioning

Every methodological design choice is grounded in established literature:

Supervised multi-block integration RGCCA (Tenenhaus & Tenenhaus, *Psychometrika*, 2011); SGCCA / variable selection (Tenenhaus *et al.*, *Biostatistics*, 2014); block relaxation (De Leeuw, 1994; Tenenhaus *et al.*, 2017; Girka *et al.*, 2024; thresholding: Gloaguen *et al.*, 2017).

Cooperative learning Multi-view cooperative learning (Ding, Li, Narasimhan & Tibshirani, *PNAS*, 2022).

Regularization & GLM Lasso / Elastic-Net with proximal gradient descent; multinomial logistic regression (GLM, logit link); class weighting via `class_weight="balanced"`.

Stability & validation Bootstrap (Efron, 1979); stability selection (Meinshausen & Bühlmann, 2010); permutation testing (Ojala & Garriga, 2010).

Direct lineage: identical dataset (IGR pediatric gliomas) and experimental setting as the RGCCA case study; SGCCA + LDA **reproduces** the published benchmark (CV $\simeq 0.826 \pm 0.136$, test $\simeq 91.7\%$).

Contributions: rigorous cross-family benchmark; algorithmic diagnosis of Cooperative divergence for $\rho > 0$; analytical breakdown of OvR vs. multinomial for rare class `midl`; stability analysis beyond point accuracy.

Comprehensive bibliography provided in the full report.

Conclusion

Key Takeaways

- **Challenging setting**: $p \gg n$ + multi-omics + rare class `midl`;
- **SGCCA + LDA**: selected model: highest overall accuracy (test $\simeq 0.924$) *and* highest stability, with `midl` recall of 2/3;
- **Sparse and interpretable** model ($\simeq$ 68 genes, 11 CGH regions);
- Signal rigorously validated via permutation test ($p \simeq 0.002$).

Future Directions:

- Validation on external patient cohorts; biological pathway analysis of stable core genes;
- Multi-component SGCCA; integration of a 3rd block (clinical covariates).

Thank you for your attention!